

$$\begin{array}{l}
 |\downarrow\rangle \\
 \\
 |\uparrow\rangle
 \end{array}
 \quad
 \longrightarrow
 \quad
 \begin{array}{l}
 f_{\uparrow}^{\dagger} |0\rangle = |\uparrow\rangle \\
 f_{\downarrow}^{\dagger} |0\rangle = |\downarrow\rangle \\
 \\
 f_{\uparrow}^{\dagger} f_{\downarrow}^{\dagger} |0\rangle = |\uparrow\downarrow\rangle \\
 |0\rangle = |\emptyset\rangle
 \end{array}
 \quad
 \left. \vphantom{\begin{array}{l} f_{\uparrow}^{\dagger} |0\rangle = |\uparrow\rangle \\ f_{\downarrow}^{\dagger} |0\rangle = |\downarrow\rangle \end{array}} \right\} \mathcal{H}_{\text{phys}}
 \quad
 \left. \vphantom{\begin{array}{l} f_{\uparrow}^{\dagger} f_{\downarrow}^{\dagger} |0\rangle = |\uparrow\downarrow\rangle \\ |0\rangle = |\emptyset\rangle \end{array}} \right\} \mathcal{H}_{\text{aux}}$$